

Flood Alerts Work. People Still Die.

An internal briefing on the gap between warning and action – and what we can do about it.



The Alert System Is Solved



What Already Exists

India's flood warning infrastructure is more capable than most people realize. The warning *generation* problem is largely solved:

ASDMA / FLEWS

Generates river-level forecasts for all major waterways in Assam

Sachet App (NDMA)

Pushes alerts nationally across **12 languages** to registered users

Cell Broadcast System

Sends loud pop-up alerts directly to phones – no app required

Upper Assam, July 2024: Where It Broke

Coverage Gap

Cell Broadcast was only a **Kamrup Metro pilot** – it never reached the hardest-hit districts in Upper Assam

Language Barrier

Alerts that did go out were **English-only at first** – a real barrier for Assamese and local-language speakers in rural communities

The Unanswered Question

People got warned. **Then what?** No clear guidance on where to go, which roads were open, or where shelter was available



The Real Gap: Alert to Action

**Warning sent
≠ People safe**

Generating and sending a warning is solved. **Everything after the alert fires is broken.**

Getting people to safety requires answers to questions the alert doesn't address:

- Which roads are still passable?
- Where is the nearest open shelter?
- How do I leave if my phone has no signal?
- What do I do if I can't read the alert?

❏ The bottleneck is not forecasting. It is the human and logistical chain between a warning and a person reaching safety.

Five Paths Forward

Each path targets a distinct documented failure from July 2024. These are options to evaluate – not a ranked recommendation.



Last-Mile Relay

When towers go down, relay via **SMS chains, community radio, or mesh networks** (Bluetooth / LoRa) so the alert reaches people even without cellular coverage



Hyperlocal Risk

Street-by-street or village-by-village flood forecasts – not just district-level river readings that average out local danger



Voice in Local Languages

IVR calls in **Assamese, Bodo, Khasi, Garo, Mizo** – reaching people without smartphones or with low literacy who are missed by app-based alerts

Five Paths Forward (continued)

1

Two-Way Reporting

Village heads and trained volunteers report ground conditions – cracking hillsides, rising banks, blocked roads – **back into the system** to close the information loop

2

What-To-Do-Next Layer

Real-time rerouting around blocked roads **plus shelter-capacity tracking** – so an alert tells you not just that danger is coming, but exactly where to go





Why This Matters Now

The Gap Is Real and Recent

Upper Assam July 2024 is not a historical case study – it is a live example of the same failure pattern repeating in a system we assumed was working

Tech Alone Is Not the Fix

Experts are consistent: the solution pairs **forecasting with community participation and local knowledge** – not just better hardware or wider app coverage

Each Path Maps to a Documented Failure

Every solution path on the previous slides addresses a specific, traceable breakdown from July – not a hypothetical gap we're guessing at

What We Need to Decide



Which Gap First?

Do we start with last-mile connectivity, hyperlocal forecasting, language access, two-way reporting, or the what-to-do-next layer?



Who Do We Partner With?

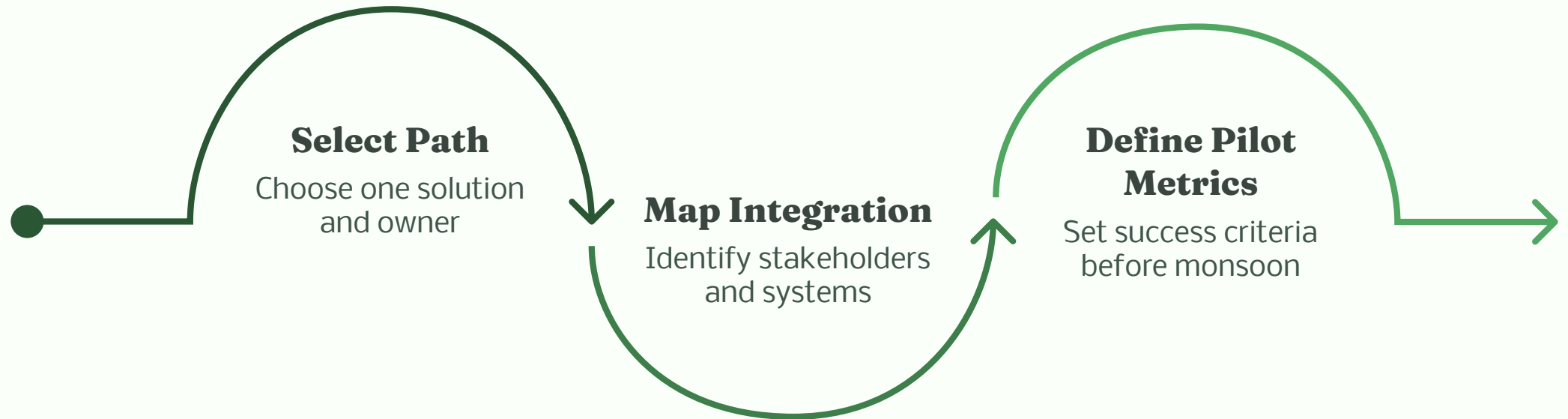
ASDMA? NDMA? Local NGOs already embedded in flood-affected districts? Existing community radio networks?



Pilot Scope and Timeline?

Which district or corridor do we test in first? What does a credible pilot look like before the next monsoon season?

Next Steps



The goal leaving this meeting: one path chosen, one owner per step, and a clear timeline to pilot before the 2025 monsoon window opens.